PHYSICS ONLY

4 MOMENTUM

Momentum is possessed by masses in motion – it is calculated by multiplying the mass of an object by its velocity. In this chapter you will learn that, when objects speed up or slow down, the rate of change of momentum is proportional to the force causing the change. You will also see that momentum is conserved in collisions and explosions.



▲ Figure 4.1 Newton's cradle

LEARNING OBJECTIVES

- Know and use the relationship between momentum, mass and velocity:

  momentum = mass × velocity

  $p = m \times v$
- Use the idea of momentum to explain safety features
- Use the conservation of momentum to calculate the mass, velocity or momentum of objects
- Use the relationship between force, change in momentum and time taken:

  $$\text{force} = \frac{\text{change in momentum}}{\text{time taken}}$$

  $$F = \frac{(mv - mu)}{t}$$
- Demonstrate an understanding of Newton's third law

KEY POINT

Do not confuse momentum, a property of moving masses, with moment, the turning effect of a force.

Newton's cradle is an entertaining toy but it also demonstrates a physics conservation law. When one of the balls is drawn back a short way and released, it swings and collides with the remaining group of balls. After the impact the ball at the opposite end springs away and swings out as far as the first ball was drawn back to start with. If two balls are drawn back and released then two balls will move away at the opposite end as the **collision** occurs. The moving ball has momentum, and momentum is conserved in collisions. This chapter is about momentum and what is meant by conservation of momentum.

MOMENTUM

We talk about objects 'gaining momentum' in everyday speech. When we say this we are usually trying to get across an idea of something becoming more

difficult to stop. Sometimes we use the word in a way that is very close to the way it is defined in physics – for example, if a car starts rolling down a hill we might say that the car is 'gaining momentum' as it speeds up.

In physics, momentum is a quantity possessed by masses in motion. Momentum is a measure of how difficult it is to stop something that is moving. We calculate the momentum of a moving object using the equation:

momentum, p (kg m/s) = mass, m (kg) × velocity, v (m/s)

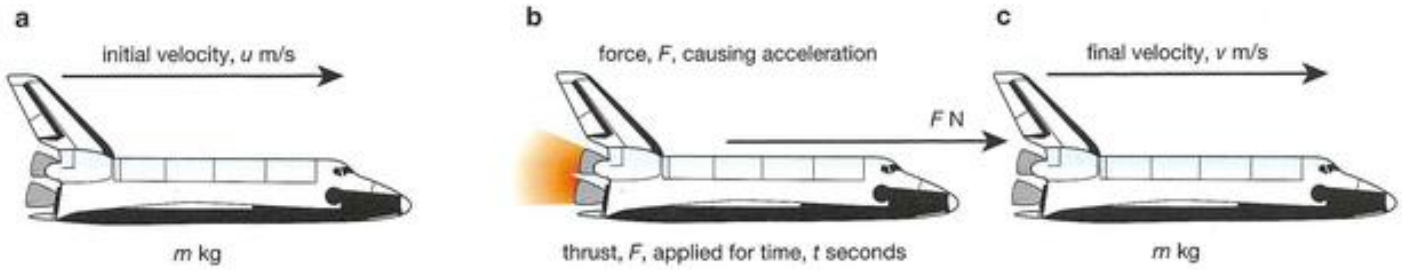
$$p = m \times v$$

Momentum is a vector quantity and is measured in kilogram metres per second (kg m/s) – provided that mass in the above equation is measured in kg and velocity in m/s.

You can see from the equation that the more mass an object has, the more momentum it will have when moving. The momentum of a moving object also increases with its speed.

MOMENTUM AND ACCELERATION

We have already discussed the relationship $F = m \times a$ (see page 30). This relationship was first discussed by Sir Isaac Newton (1642–1727). A more precise statement of Newton's discovery would be that when an unbalanced force acts on an object it causes a change in the momentum of the object in the direction of the unbalanced force. Newton discovered that the rate of change of momentum of an object is proportional to the force applied to that object. This means that if you double the force acting on an object its momentum will change twice as quickly. Figure 4.2 shows the effect of the thrust force on the velocity of a space shuttle, and, therefore, on its momentum.



▲ Figure 4.2 The thrust of the rocket motor makes the velocity and, therefore, the momentum of the shuttle increase.

initial momentum of object = mu

final momentum of object = mv

therefore, increase in momentum = $mv - mu$

so, rate of increase of momentum = $\frac{(mv - mu)}{t}$

As stated above, Newton identified a proportional relationship between the rate of increase of momentum and the force applied, but with the system of units we use the relationship appears as shown:

$$\text{force, } F = \frac{\text{change in momentum } (mv - mu)}{\text{time taken, } t}$$

$$F = \frac{mv - mu}{t}$$

If you look at this equation you will notice it can be rearranged to give the more familiar equation $F = ma$:

$$F = \frac{mv - mu}{t}$$

$$= \frac{m(v - u)}{t}$$

since $\frac{(v - u)}{t} = a$

then $F = ma$

The rearrangement is possible because we have assumed that the mass of the object involved is constant. This will not always be the case in real situations – for example, when a space shuttle is launched the mass of the rocket changes continuously as fuel is burned and rocket stages are jettisoned. Rocket scientists also have to include air resistance in their calculations. However, you will not meet problems like this at International GCSE level.

EXAMPLE 1

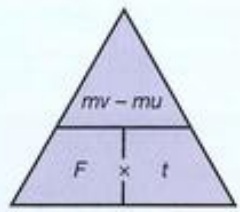
The first stage of the type of rocket used in Moon missions provides an unbalanced upward (away from the Earth) force of 30 MN and burns for 2.5 minutes.

a Calculate the increase in the rocket's momentum that results.

b If the rocket has a mass of 3000 tonnes what is the velocity of the rocket after the first stage has completed its 'burn'?

(Take care to convert units into N, kg and s. 1 MN = 10^6 N; 1 tonne = 1000 kg)

a The equation $F = \frac{(mv - mu)}{t}$ is another that can be rearranged using the triangle.



▲ Figure 4.3 Rearranging the change in momentum = $f \times t$ equation using the triangle method

We want the change in momentum (at the top of the triangle) so it is at the bottom of the triangle:

$$F \times t = (3 \times 10^7 \text{ N}) \times (2.5 \times 60 \text{ s})$$

increase in momentum is 4.5×10^9 kg m/s

b The rocket starts from rest, $u = 0$, therefore the initial momentum (mu) is 0. This means that the increase in momentum is $mv = 4.5 \times 10^9$ kg m/s. Divide this by mass, $m = 3 \times 10^6$ kg, to give:

$$v = \frac{4.5 \times 10^9 \text{ kg m/s}}{3 \times 10^6 \text{ kg}}$$

$$= 1.5 \times 10^3 \text{ m/s}$$

As we have seen above this has been calculated with the mass of the rocket staying the same. In a real launch fuel is used up so the mass of the rocket will decrease and the calculation would be harder.

MOMENTUM AND COLLISIONS

The total momentum of objects that collide remains the same:

momentum before the collision = momentum after the collision

You will need to use this to work out what happens to objects that collide when they are moving along the same straight line.

EXTENSION WORK

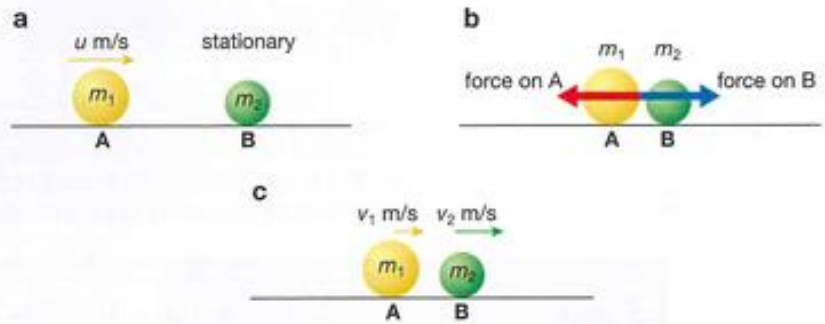
You will not need to learn the following explanation for the exam, but you do need to understand Newton's third law. You will see this again later in this chapter.

We can express the above equation in terms of momentum change, as follows:

force × time = increase in momentum

This simply says that a bigger force applied to an object for a longer time will result in a greater change in the momentum of the object.

Consider what happens when two balls collide, as shown in Figure 4.4.



▲ Figure 4.4 **a** Moving ball A, mass m_1, rolls towards stationary ball B, mass m_2; **b** During the impact each ball exerts a force on the other – equal in size and opposite in direction; **c** The balls after the collision.

EXTENSION WORK

During the time the two balls are in contact each exerts a force on the other (Newton's third law about action and reaction, see page 46). The forces act in opposite directions and obviously act for the same amount of time. This means that $F \times t$ for each is the same size, but opposite in direction.

KEY POINT

The term impulse is used for the product (force × time).

The increase in momentum of ball B is exactly balanced by the decrease in momentum of ball A, so the total momentum of the two balls is unchanged before and after the collision – momentum is conserved.

EXTENSION WORK

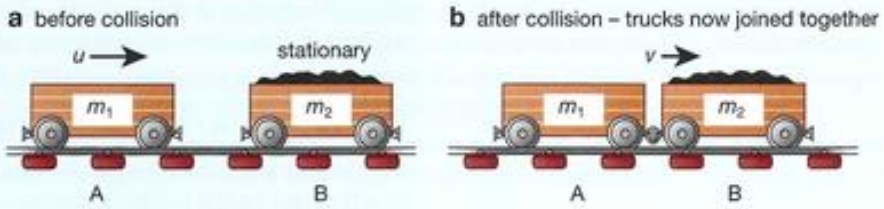
In any system, momentum is always conserved provided no external forces act on the system. This means that when two snooker balls collide the momentum of the balls is conserved if no friction forces act on them. The presence of friction means that the balls will eventually slow down and stop, thus ending up with no momentum. Although the balls have 'lost' momentum something else will, inevitably, have gained an equal amount of momentum! As the balls are slowed by the friction of the snooker table they, in turn, cause a friction force to act on the table. The table gains some momentum. However, the large mass of the table means that the effect is unnoticeable.

EXAMPLE 2

A railway truck with a mass of 5000 kg rolling at 3 m/s collides with a stationary truck of mass 10 000 kg. The trucks join together. At what speed do they move after the collision?

We shall assume that friction forces are small enough to ignore, so we can apply the law of conservation of momentum:

momentum before the collision = momentum after the collision



▲ Figure 4.5 Railway trucks in collision

so, momentum of A before collision + momentum of B before collision = momentum of A and B moving together after collision

$$(m_1 \times u) + (m_2 \times 0 \text{ m/s}) = (m_1 + m_2) \times v$$

where m_1 is the mass of truck A, u is its velocity before the collision, m_2 is the mass of truck B (at rest before the collision so its velocity is 0 m/s), and v is the velocity of the two trucks after the collision.

Substituting these values gives:

$$(5000 \text{ kg} \times 3 \text{ m/s}) + (10\,000 \text{ kg} \times 0 \text{ m/s}) = (5000 \text{ kg} + 10\,000 \text{ kg}) \times v$$

so $$v = \frac{15\,000 \text{ kg m/s}}{15\,000 \text{ kg}} = 1 \text{ m/s}$$

After the collision the trucks move with a velocity of 1 m/s in the same direction that the original truck was travelling.

EXPLOSIONS



▲ Figure 4.6 A shuttle launch

The conservation of momentum principle can be applied to explosions. An explosion involves a release of energy causing things to fly apart. The momentum before and after the explosion is unchanged, though there will be a huge increase in movement energy. A simple demonstration of a safe 'explosion' is shown in Figure 4.6. You might think that the rocket in this photo is taking off because the rocket motor is pushing against the ground, but rockets work in space where there is nothing to push against.

Rocket motors use the principle of conservation of momentum to propel spacecraft through space. They produce a continuous, controlled explosion that forces large amounts of fast-moving gases (produced by the fuel burning) out of the back of the rocket. The spacecraft gains an equal amount of momentum in the opposite direction to that of the moving exhaust gases.

You can see the same effect if you blow up a balloon and release it without tying up the end!

CAR SAFETY

In Example 1 on page 31 you saw that the force on a person in a car crash can be very large. In that example, the force was worked out from the deceleration in a crash. You can also work out the force in a crash using the equation for momentum.

Steep roads often have escape lanes filled with deep, soft sand. The soft sand slows heavy lorries that are out of control slowly – by making time, t, for the lorry to stop longer, the force, F, slowing the lorry is smaller and the driver is less likely to suffer serious injury,

EXAMPLE 3

A car travelling at 20 m/s collides with a stationary lorry and is brought to rest in just 0.02 s. A woman in the car has a mass of 50 kg. She experiences the same deceleration when she comes into contact with a hard surface in the car (such as the dashboard or the windscreen). What force does the person experience?

$$\text{force} = \frac{\text{change in momentum}}{\text{time}}$$

$$= \frac{(50\text{ kg} \times 20\text{ m/s}) - (50\text{ kg} \times 0\text{ m/s})}{0.02\text{ s}}$$

$$= 50\,000\text{ N}$$



▲ Figure 4.7 Parts of cars are designed to crumple.

Cars are now designed with various safety features that increase the time over which the car's momentum changes in an accident. Figure 4.7 shows the safety features of a car being tested. The car has a rigid passenger cell or compartment with crumple zones in front and behind. The crumple zones, as the name suggests, collapse during a collision and increase the time during which the car is decelerating. For instance, if the deceleration time in Example 3 above is increased from 0.02 s to 1 s, then the impact causes a much smaller force of just 1000 N to act on the passenger, greatly increasing their chances of survival.

Crumple zones are just one of the safety features now used in modern cars to protect the passengers in an accident. They only work if the passengers are wearing seat belts so that the reduced deceleration applies to their bodies too. Without seat belts, the passengers will continue moving forward until they come into contact with some part of the car or with a passenger in front. If they hit something that does not crumple they will be brought to rest in a very short time, which, as we have seen in Example 3, means a large deceleration and, therefore, a large force acting on them.

NEWTON'S LAWS OF MOTION

EXTENSION WORK

Scientists use the word 'law' very cautiously. Only when a hypothesis (idea) has been tested many times independently by careful experiment is it raised to the status of a 'law'. Einstein showed that in special situations Newton's laws break down, but they are still accurate enough to predict the way objects respond to forces with a high degree of accuracy.

EXTENSION WORK

Sir Isaac Newton lived from 1642 to 1727. He made many famous discoveries and some important observations about how forces affect the way objects move. The first observation, called Newton's first law, was:

> *Things don't speed up, slow down or change direction unless you push (or pull) them.*

Newton didn't put it quite like that, of course! He said that a body would continue to move in a straight line at a steady speed unless a resultant (unbalanced) force was acting on the body. If the forces acting on an object are balanced then a stationary object stays in one place, and a moving object continues to move in just the same way as it did before the forces were applied. You have already met this idea in Chapter 3.

Newton then asked another obvious question: how does the acceleration of an object depend on the force that you apply to it? Again Newton's formal statement of the answer (Newton's second law) sounds complicated, but the basic findings are quite simple:

> *The bigger the force acting on an object, the faster the object will speed up.*

Objects with greater mass require bigger forces than those with smaller mass to make them speed up (accelerate) at the same rate.

The force referred to is the resultant force acting on the object. This idea is also covered in Chapter 3. The relationship $F = m \times a$ is a consequence of Newton's second law.

The only law you you need to learn by name is Newton's third law.

NEWTON'S THIRD LAW: ACTION AND REACTION

In simple language Newton's third law might be stated:

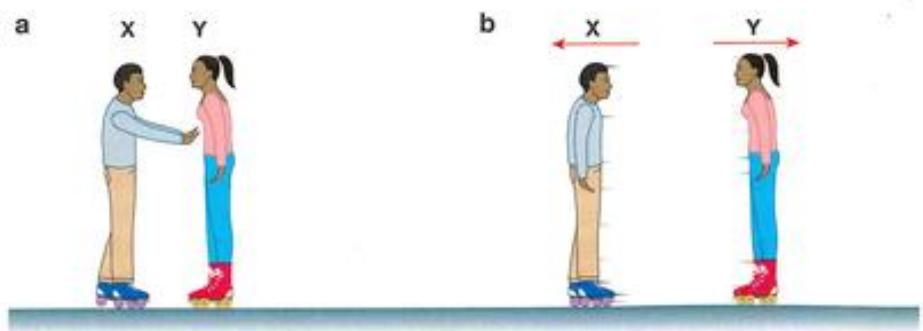
> *When you push something it pushes you back just as hard, but in the opposite direction.*

It is usually stated as:

> *For every action there is an equal and opposite reaction.*

When you sit down, your weight pushes down on the seat. The seat pushes back on you with an equal, but upward, force. An experiment to give you the idea of what this law means is shown in Figure 4.8.

In Figure 4.8, person X is clearly pushing person Y but it is not obvious that Y is pushing X back. When both X and Y move it is clear that X has been affected by a force pushing him to the left. The force felt by X is the reaction force.



▲ Figure 4.8 For every action there is an equal and opposite reaction.

Look back at Figure 2.3 on page 19. The weight of the car is pushing down on the ground. Although the weight is shown coming from the centre of gravity of the car, it acts through the wheels pressing down on the ground. The total reaction force from the ground acting upwards on the wheels is the same as the weight of the car. It is this reaction force that stops the car sinking into the ground.

It can sometimes be difficult to sort out action and reaction forces from balanced forces. Balanced forces act in opposite directions on the same object. Action and reaction forces also act in opposite directions, but are always acting on different objects.

CHAPTER QUESTIONS

More questions on momentum can be found at the end of Unit 1 on page 55.

SKILLS PROBLEM SOLVING

1 Work out, giving your answers in kg m/s, the momentum of the following moving objects:

- a a bowling ball of mass 6 kg travelling at 8 m/s
- b a ship of mass 50 000 kg travelling at 3 m/s
- c a tennis ball of mass 60 g travelling at 180 km/h.

2 An air rifle pellet of mass 2 g is fired into a block of modelling clay mounted on a model railway truck. The truck and modelling clay have a mass of 0.1 kg. The truck moves off after the pellet hits the modelling clay with an initial velocity of 0.8 m/s.

- a Calculate the momentum of the modelling clay and truck just after the collision.
- b State the momentum of the pellet just before it hits the modelling clay.
- c Use your answers to a and b to calculate the velocity of the pellet just before it hits the modelling clay.

SKILLS REASONING

- d State any assumptions you made in this calculation.

SKILLS PROBLEM SOLVING

3 A rocket of mass 1200 kg is travelling at 2000 m/s. It fires its engine for 1 minute. The forward thrust provided by the rocket engines is 10 kN (10 000 N).

- a Use increase in momentum = $F \times t$ to calculate the increase in momentum of the rocket.
- b Use your answer to a to calculate the increase in velocity of the rocket and its new velocity after firing the engines.

SKILLS CRITICAL THINKING

4 Air bags are a safety feature fitted to all modern cars. In the event of sudden braking an air bag is rapidly inflated with gas in front of the driver or passenger.

- a Does the air bag remain inflated?

SKILLS REASONING

- b Explain how the air bag protects the driver from more serious injury.

END OF PHYSICS ONLY